

Amit Schechter

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Education

Massachusetts Institute of Technology (MIT) <i>Doctor of Philosophy (PhD) candidate, Computer Science</i> • Research Topics: Machine Learning Robustness, Generative Models, Computer Vision, Vision Language Models • Advisor: Boris Katz	2024 – Present
Massachusetts Institute of Technology (MIT) <i>Master of Science (SM), Computer Science</i> • GPA: 4.6/5.0 • Advisor: Tommi S. Jaakkola • <i>Master's Thesis</i>: Methods for Enhancing Robustness and Generalization in Machine Learning	2019 – 2024
Stanford University <i>Bachelor of Science (BS), Computer Science</i> • GPA: 4.0/4.0, graduated with distinction • Artificial Intelligence concentration	2015 – 2019

Honors and Awards

MIT Graduate Presidential Fellowship , MIT, MA	2020
Bachelor's Degree with Distinction (highest award: top 15%) , Stanford, CA	2019
President's Award for academic excellence , Stanford, CA	2019
Apple Best Software Competition Winner , Stanford, CA	2018
Tau Beta Pi Engineering Honor Society , Stanford, CA	2018
Phi Beta Kappa Honor Society , Stanford, CA	2018
Stanford Excellence Fellowship , Stanford, CA	2015-2019

Experience

MIT Computer Science & Artificial Intelligence Laboratory , Cambridge, MA <i>Graduate Research Assistant. Advisors: Prof. Tommi S. Jaakkola, Prof. Boris Katz</i> • Generative modeling, computer vision: Concept erasure via attention redirection, and structured representation extraction in generative video models. • Out-of-distribution robustness and generalization: continuous group DRO and invariant decision trees	2019-Present
Stanford Machine Learning Research Group , Stanford, CA <i>Undergraduate Researcher, machine learning for healthcare. Advisor: Prof. Andrew Ng</i> • Used deep learning for chest x-ray abnormality detection; model outperformed radiologists and top experts • Collaborated with MSK hospital to develop a tool for heart failure detection and prediction	Apr 2018 – June 2019
Google, Flights (QPX) team , Cambridge, MA <i>Software Engineering Intern, generative models research</i> • Researched models for generating discrete data for augmenting Google Flight clicks • Techniques & technologies: Generative Adversarial Networks (GANs), discrete data representation, TensorFlow	June 2018 – Sept 2018
Goldman Sachs - Equities Engineering , New York, NY <i>Software Engineering Intern</i> • Developed a web app to package trades, used by multiple teams for margin calculation (ReactJS, Redux)	Jan 2018 – Mar 2018
Google, YouTube Ads Infrastructure team , Mountain View, CA <i>Engineering Practicum Intern (Software Engineering)</i>	June 2017 – Sept 2017

- Developed a configurable production tool to rapidly filter YouTube ads for sudden unusual circumstances.
Reduced reaction time by over 90%

Stanford MobiSocial Lab, Stanford, CA
Undergraduate Research Assistant

June 2016 – Aug. 2016

- SwipeVR, A fast virtual keyboard for VR applications (C++ , Android dev, VR-game design, product research)

Publications

Concept Erasure via Attention Redirection, *CVPR 2026 Findings*

Amit Schechter, Rinon Gal, Ofir Kedem, Gal Chechik, Daniel Cohen-Or

CheXaid: deep learning assistance for physician diagnosis of tuberculosis using chest x-rays in patients with HIV, npj Digital Medicine

Pranav Rajpurkar*, Chloe O'Connell*, **Amit Schechter***, Nishit Asnani, Jason Li, Amirhossein Kiani, Robyn L Ball
Marc Mendelson, Gary Maartens, Daniël J van Hoving, Rulan Griesel, Andrew Y Ng, Tom H Boyles, Matthew P Lungren
10.1038/s41746-020-00322-2

Additional Research Projects

Weighted Group Distributionally Robust Optimization

- Developed a method to generalize Group DRO. Improved robustness with noisy, or uncertain group assignments

Robust Decision Trees

- Developed methods for improving robustness of decision trees using invariant risk minimization
- Showed theoretically and experimentally that the proposed method improves robustness to distribution shifts

Teaching

Modeling with Machine Learning for Computer Science, MIT

Spring 2023, 2024, 2025

Teaching Assistant with Prof. Tommi Jaakkola. Assisted in course development.

Graduate Machine Learning (6.790), MIT

Fall 2024, 2025

Teaching Assistant with Prof. Leslie Kaelbling, Tamara Broderick, Guy Bresler

AI, Decision Making, and Society, MIT

Fall 2022

Teaching Assistant with Prof. Aleksander Madry

Modeling with Machine Learning: from Algorithms to Applications, MIT

Spring 2021

Teaching Assistant with Prof. Tommi Jaakkola and Prof. Regina Barzilay

Programming Methodology, Programming Abstractions, Stanford

Jan 2017 – Sept 2017

Undergrad TA Stanford Computer Science - CS 198 program (106A and 106B)

Activities

we:fm – Music Sharing App Developer – Winner of Apple Software Competition

Designed and developed a music sharing app. Winner of Apple's best app in the Stanford software fair

MIT Graduate Women in Course 6 Board Member

Board Member in GW6, a group that builds community and supports graduate women in computer science at MIT

Mentor, Girls Teaching Girls to Code (Stanford outreach program)

Mentored ~200 high school girls in coding education program, focused on computer graphics

Skills

Programming: TensorFlow, PyTorch, Python, C++ , C, Java, JavaScript

Tools & Platforms: Android and iOS app development, HTML, React

Languages: Hebrew - Native, English - Fluent